The University of Zambia

in association with

ZCAS University

BFS322 TREASURY MANAGEMENT

TEST

22 October 2018

08:00 – 11:00 HOURS

TIME ALLOWED: THREE HOURS

INSTRUCTIONS:

1. ALL questions in this paper are COMPULSORY and must be attempted
2. This examination paper carries a total of 100 marks.
3. Candidates must **not turn this page** until the invigilator tells them to do so.

ALL QUESTIONS ARE COMPULSORY AND MUST BE ATTEMPTED

Q1

The Bank of Zambia (BOZ) at its Monitory Policy Committee (MPC) meeting held on 20-21st August 2018, decided to maintain the Policy Rate at 9.75% despite the inflation rate increasing slightly to 9.75% in June 2018.

i. Identify the department within BOZ that is responsible for the implementation of monitory policy and discuss the role it plays when it comes to treasury management.

(25 marks)

ii. Critically analyse the reasons why the MPC decided to maintain the Policy Rate at 9.75% and discuss the risks to BOZ economic outlook.

(15 marks)

Q2

Nkongole Enterprises Limited is a manufacturer of exotic furniture. You have been recently appointed as the company Treasurer. The company has been facing some working capital challenges recently, which has led it to rely on its overdraft facility with the bank on which it pays 20% p.a. in interest.

One of your first tasks as a Treasurer is to examine the management accounts of the company and you have been given the following information:

	2018	**2017**	**2016**
Cash	K0	K0	K5,000
Debtor days	90	60	30
Creditor days	120	100	45
Stock turnover per month	2	2.5	4

Required

i. Based on the above figures alone, advise management on the causes of the working capital problems that the company is facing. **(5 marks)**

ii. Suggest one method that management can implement in order to manage its stock and give the benefits of this method. **(5 marks)**

iii. In order to manage generate cash, the company has offered its customers a cash discount of 5% on an outstanding invoice of K1000 if the customer settles the invoice in 10 days. The company has 30 days credit terms with its customer. What will be the impact of this cash offer on the company assuming that the company faces a cost of funds of 10%? **(10 marks)**

Q3

Treasury is one of the key departments within an organisation. The role of the treasurer encompasses many tasks which are key to the financial wellbeing of any organisation. The treasury role has evolved over the years from the simple cash management functions to more complex risk management functions. Discuss the evolving role of treasury and give a historic perspective on the events that have had an impact on treasury. **(20 marks)**

Q4

Cash forecasting assists Treasury in planning an organisation's cash management. Given the adage goes 'cash is king!' the importance of cash management to an organisation cannot be over emphasised.

Outline the purposes for cash management and identify any three (3) issues that the Treasurer may consider when preparing cash forecasts. **(20 marks)**

END OF TEST

MARKING GUIDE/KEY

Q1

(i)

Financial Markets Department (5)

- The Financial Markets department is the department at BOZ that plays the treasury role
- Implements monetary policy and act as a fiscal agent of the government
- Plays critical role in contributing to BOZ objective of achieving price stability

Implementation of Monetary Policy (5)

- MPC sets the policy rate which indicates BOZ monetary stance
- Achieved by department undertaking OMO to either supply or withdraw liquidity using various financial instruments
- A stable liquidity position is maintained in financial system in line with desired level of interest rates
- Influencing short term interest rates at appropriate levels assists in achieving price and financial system stability

Administering the OLF (5)

- A liquidity financing facility which commercial banks can access in need of liquidity from BOZ at day end
- OLF complements other refinancing facilities such as TB Rediscount Facility and Intraday Credit Facility
- Primary purpose of OLF is to enhance operations of money markets by assuring commercial banks of liquidity assistance from BOZ

Fiscal Agent of Government (5)

- Executed through issuance and redemption of GRZ securities
- Conducts primary auctions of GRZ securities, redemptions and interest payments as well as maintaining GRZ securities databases

- Advises and provides domestic debt projections and estimations of domestic debt service to MOF for annual national budget purposes

Managing Foreign Exchange Reserves (5)

- Assets denominated in foreign currency held by monetary authorities
- Consist of cash, financial instruments, gold and SDRs
- Department undertakes foreign exchange dealings that enables Bank to respond to core function of price and financial system stability
- Zambia has flexible exchange rate system
- Exchange rate policy management aimed at ensuring relative stability in exchange rate
- Executed by interventions in FX market

(ii)

Reasons why BOZ decided to maintain Policy Rate at 9.75%: (10)

- Inflation remained within the target range of 6-8% and projected to remain anchored within this range
- Commercial Bank nominal average lending rates increased marginally
- Subdued economic growth due to high lending rates and reduced domestic credit
- Need to enhance the stability of the financial sector

Risks to BOZ economic outlook: (5)

- Rising inflation above target corridor due to depreciation of the kwacha
- Worsening fiscal deficits
- Slow down in global economic growth due to Brexit, trade tensions etc

Q2

(i) The working capital problems could be caused by the following: (1) for each point

- The company has run down its cash over the years
- This could be due to the fact that the cash conversion cycle has been lengthening over the years
- Its customers are taking longer to pay, as revealed by the worsening debtor days. This has the effect of tying up cash in the receivables
- It is taking the company longer to turn over its stock. This means that cash is increasingly tied up in unsold stock

- As a result of the worsening cash position of the company, it finds itself unable to meet its commitments as revealed by the worsening creditor days position.

(ii) Just in Time Method (1) for each point

- Management can implement a Just In Time method
- JIT purchasing means that materials or components are received from the supplier just in time to use them.
- Benefits of JIT purchasing:
  i. Raw material stock reduction;
  ii. Frequent deliveries with smaller orders from a smaller number of suppliers;
  iii. Long term contracts with suppliers
  iv. Quality assurance, with supplier becoming responsible for inspection of goods supplied
- Objectives of JIT production are to obtain low-cost, high quality, on time production, to order,
- Achieved by minimising idle stock levels between successive processes and thereby minimising idle equipment, facilities and workers

(iii)

- To determine the present value (PV) of receiving the discounted payment on day 10, begin by calculating the discounted payment amount as follows:

  PV = FV/(1+r)

- And the interest rate (i) for the discounted period can be calculated as:

  i= days in discount period x annual interest/365

  = 10 x 10%/365 = 0.27% (2)

- The two can be combined to calculate the PV of the discounted payment as:

- $Pv_{day\ 10}$ = K950/(1+0.27%)= K947.44 (2)

- Alternatively the PV of receiving the full payment after 30 days is calculated as:

- $Pv_{day\ 30}$ = K1000/(1+0.82%) = K991.87 (2)
  - The seller can now calculate the NPV of the discount as follows:

- NPV= PV_{day10} – PV_{day30} = K947.44-K991.87 = -K44.43 (2)
- From results, the seller knows its net disadvantage is K44.43 for every K1000 of credit sales if the buyer takes the discount and pays by day 10 rather than wait to pay the full amount on day 30. (2)

Q3

Historical Events that have led to evolving treasury (4 marks each)

i) Increased volatility in exchange and interest rates

- Break down of Bretton Woods system of fixed exchange rates
- Inflationary period following oil shocks in 1970s and inflationary monetary policies leading to high levels of interest rates in developing countries

ii) Stock Market volatility

- Due to high inflation in 1970s
- Stock market crash in 1970s
- Stock market growth in 1990s followed by crush in 2000 (dotcom bubble)

iii) Regulation of banks in 1970s led to reduced competition amongst banks

- De-regulation in 1980s led to more competition in financial sector (big bang in UK)

iv) Increased levels of securitization in 1990s

- Options and futures

v) Developments in technology

- Increasing use of PCs
- Increase in processing power
- Increase in memory capacity
- Evolution of the internet

Q4

Purpose of cash forecasting (2 marks for each point)

1. Managing liquidity
   Forecasting a company's net cash position to identify potential cash excesses or shortages is essential for scheduling investment decisions and anticipating borrowing requirements to meet daily obligations

2. controlling financial activities
   Helps identify problems such as unanticipated inventory changes, delays in A/R collection, mistiming of payments and fraud and embezzlement.

3. Meeting strategic objectives
   Used to project future funding requirements and make operating decisions to support a company's strategic plan

4. Budgeting capital
   Helps in evaluating proposed capital expenditures

5. Managing costs
   Used to help minimise excess bank balances, reduce short term borrowing costs, and increase short term investment income

6. Managing currency exposure
   Used to assess the degree of FX exposure

7. Complying with regulatory requirements

- Part of internal control procedures needed to meet various regulatory requirements, such as minimum capital requirements or requirements for inprest tax accounts

Issues with cash forecasting (2 marks for each point)

1. Simplicity
    - Degree of accuracy is a function of the purpose of the forecast
    - Short term forecast used to manage daily cash position and has to be more precise
    - Long term forecast used to determine availability of capital may be less precise
    - Neither needs to be accurate to the last decimal

2. Collaboration and Communication
    - Treasury is dependent upon business units to provide the information and assumptions needed to create a good forecast

3. Duplication
    - Due to many uses of cash forecasts departments besides treasury often develop forecasts.
    - Information and assumptions used are similar
    - Could result in duplication of tasks
    - The results may be different leading to lack of trust in them by management